Active Directory Object extraction procedure

Brainwave GRC

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Introduction

This documentation describes how to use scripts provided by Brainwave Identity GRC to extract the objects (groups, acccounts . . .) from Active Directory environments, from single domain to the largest multiple forsest / domains configurations.

Active Directory objects extraction

The provided script `ExtractAD.ps1` allows to extract the objects managed in Active Directory such as groups, organizational units and user accounts.

Prerequisites

The extraction script is stored in `/extractors/bw_activedirectory` in the project when the installation of the add-on is done.

- ExtractAD.ps1: Extraction des objets Active Directory. Available as of version: bw_activedirectory_5.1 add-on
- ICF compatibility

This script is compatible with brainwave ICF Powershell connector

- Dependencies
  - Requires PowerShell 4.0 or greater
  - Domain user with at least “read” privilege
  - Requires .NET Framework 4.5 or greater

> **Note**: The powershell script is signed, any modification of the script will make it ineffective unless you change your local security parameters in order to authorize the execution of unsigned scripts. Please note that this modification leads to potential vulnerabilities.

Extraction command parameters

- `-filter`: An LDAP filter by default it is : `(|(objectcategory=Person)(objectclass=group))`
- `-ouFilter`: Filter on a specific organizationalUnit, should be a valid organizational Unit DN (example: `OU=Internal Users,DC=intra,DC=test,DC=local`)"

> **Note**: This filter will apply on all domains if script launched on several servers (with `--server` option)

- `-attributesfile`: File that contains the attribute names to include in the extraction according to the mode parameter (by default, `attributes.cfg`). For example: `attributes.txt` where it contains an attribute in each line for example:

  - dn
  - objectclass
  - accountexpires
  - givenname

By default this parameters contain mandatory attributes for AD facet:

```
"objectcategory","objectguid","objectsid","sidhistory","member",
"objectclass","accountexpires","badpwdcount","displayname",
"givenname","lastlogon","logoncount","lastlogontimestamp",
"mail","manager","pwdlastset","samaccountname","sn","whencreated",
"useraccountcontrol","createtimestamp","description","grouptype",
"managedby","modifytimestamp","legacyexchangedn","PrimaryGroupID"
```

- `-servers`: File that contains in each line a name of an AD server, if the file is not specified, the script will extract all the ActiveDirectory domains data in the current forest.
- `-credential`: File containing PSCredential used to extract objects in ActiveDirectory, see example to create and store PSCredential object to a file, if not set current session credential will be used.
- `-LogLevel`: This parameters allow to specify a log level. Possible values are:
  - Error
  - Warning
  - Info
  - Debug

The default value is `Error`.

- `-errAction`: Specify to the script what action to perform if an error occur The possible values are:
  - `Continue`: The script continues if an error occurs and displays it.
  - `SilentlyContinue`: The script runs without displaying errors.
  - `Stop`: The script stops if an error occurs, all errors become critical.

The default value is `Stop`

- `-printLog`: Specify to the script whether to display traces in the console. Possible values are ‘true’ and ‘false’. Default value is ‘true’
- `-port`: LDAP server port, default value is 389
- `-authType`: Authentication method to use in LDAP connection, for more details refer to https://msdn.microsoft.com/fr-fr/library/system.directoryservices.protocols.authtype(v=vs.110).aspx
- `-useSSL`: Boolean value to activate SecureSocketLayer on LDAP connection, default value is ‘False’

- `-outputDirectory`: Specify to the script where the generated data files will be stored. The default value is the current directory.
- `-logDirectory`: Specify to the script where the generated log file will be stored. The default value is the current directory.

Examples

Example 1:

The commands necessary to create the PSCredential and store it in a file (password will be encrypted by powershell API), the generated file will only work using local machine and user session where it was created.

```
$pwd=ConvertTo-SecureString --String $Password --AsPlainText -Force
$credential=New-Object --TypeName System.Management.Automation.PSCredential
    --ArgumentList $filerLogin, $pwd
```

```
$credential | Export-CliXml credential.xml
$credential = Get-Credential -UserName test\local
$credential | Export-CliXml credential.xml
```

Example 2:

A command example to extract AD data using default mode (include), default attribute list (mandatory attributes for AD facet) and default filter (extract only users and groups)

```
./ExtractAD.ps1 -servers 'servers.cfg'
```

Expected results

ActiveDirectory script generates one LDIF files by AD domain as output:

- `{DomainNetBiosName}.ldif`

The script also produces a log file. This file is named `ExtractAD{timestamp}.log` and contains details of extractions according to the -loglevel parameter.